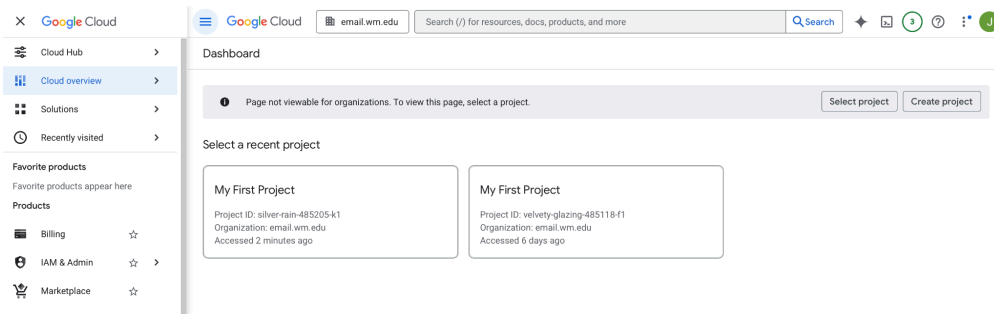
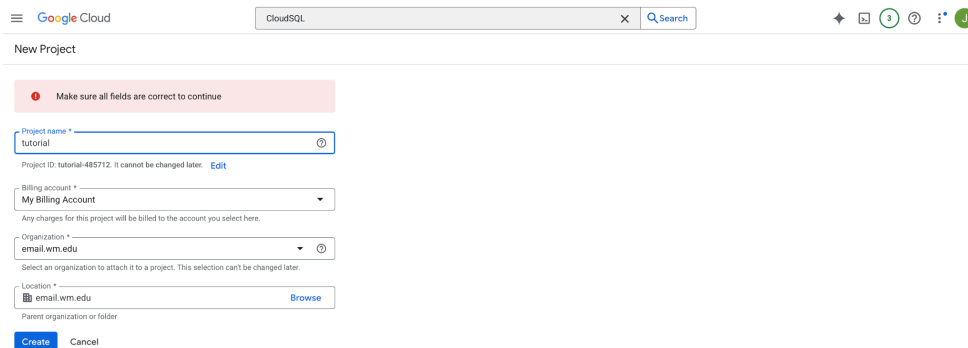


1. Log into Google Cloud Console with your school email account



a.

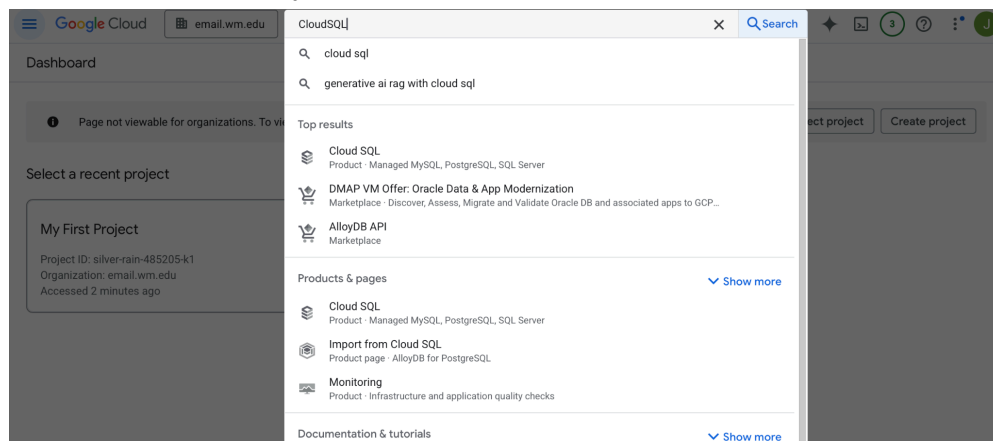
2. Select “Create project”



a.

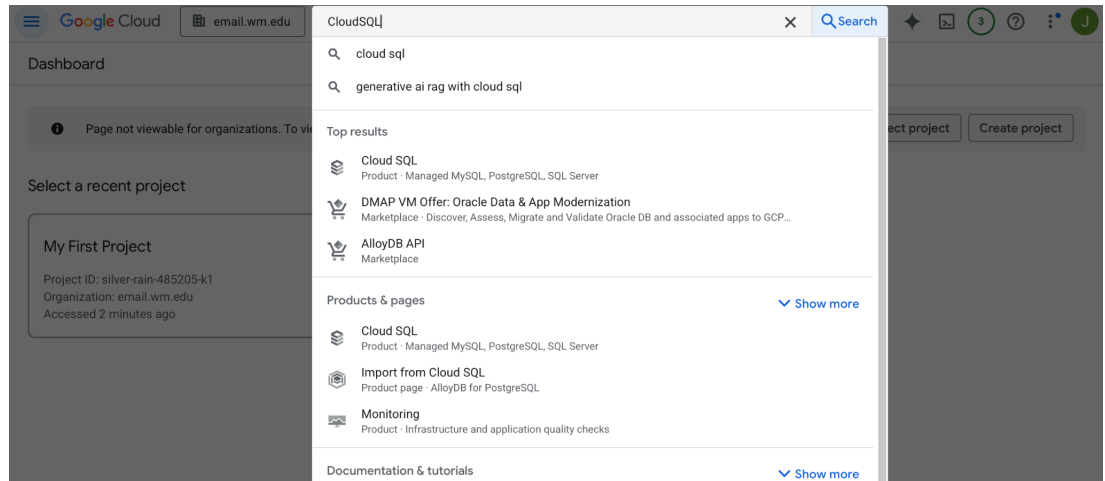
b. Fill out the Project name as your project name of interest. I put mine as “tutorial” for example here.

3. In the search bar on the top, type “CloudSQL”



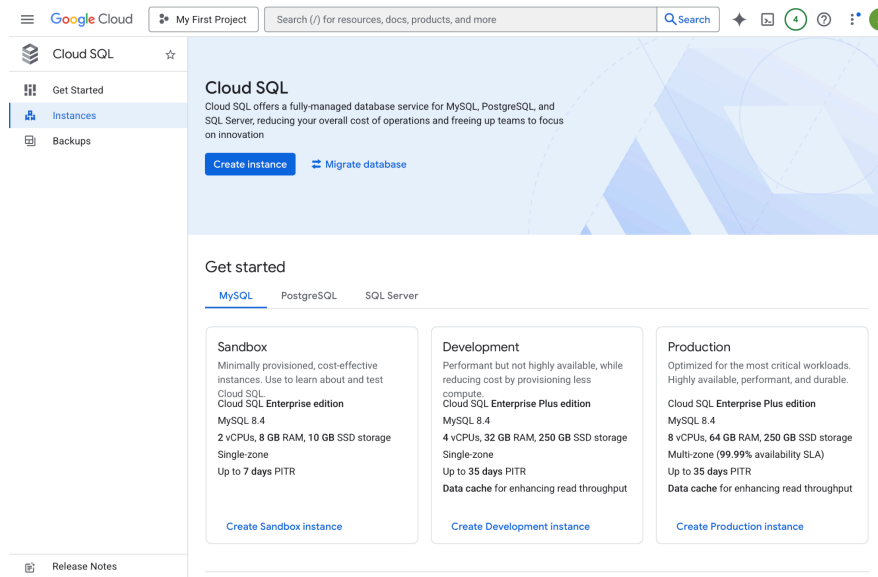
a.

4. Select the product CloudSQL from the list



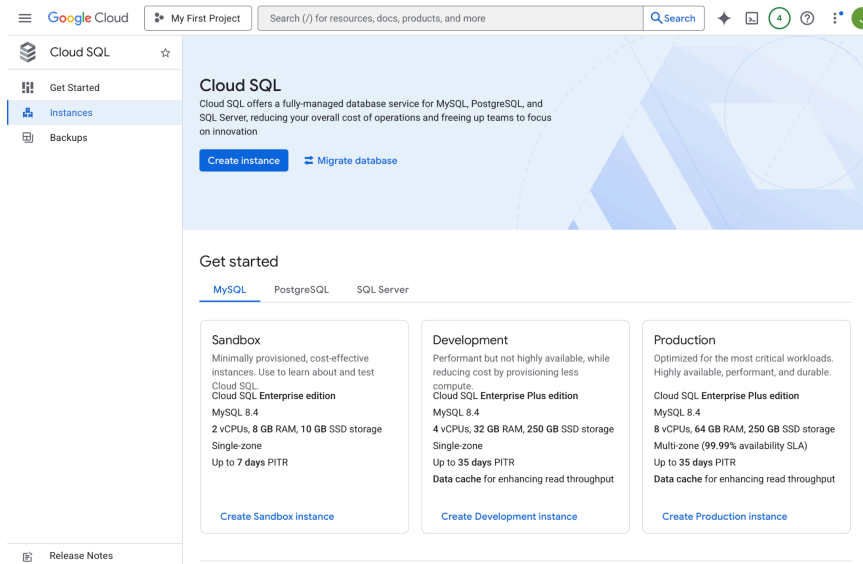
a.

5. You should be seeing this page:



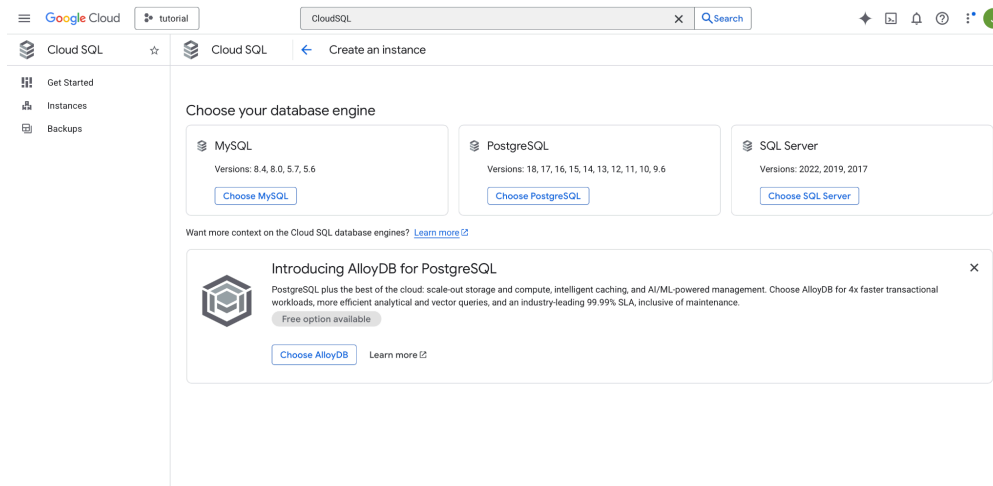
a.

6. Select "Create instance"



a.

7. In “Choose your database engine”, select “Choose MySQL”



a.

8. Configure your instance like mine below. Make sure to set your own password and remember it

Choose a Cloud SQL edition

A Cloud SQL edition determines foundational characteristics of your instance. Choose the best option for your price and performance needs. [Learn more](#)

☐ Enterprise Plus

- 99.99% availability SLA
- Sub-second planned maintenance downtime
- Near-zero downtime instance scale-up
- Performance optimized machines
- Up to 35 days point-in-time recovery window
- Up to 3x higher read throughput with data cache
- Advanced disaster recovery with easy switchback
- Support for managed connection pool

☒ Enterprise

- 99.95% availability SLA
- Less than 60 seconds planned maintenance downtime
- General purpose machines
- Up to 7 days point-in-time recovery window

Choose a preset for this edition. Presets can be customized later as needed.

Edition preset
Sandbox

a. [Compare edition presets](#)

Instance info

Database version *
MySQL 8.4

Instance ID *
tutorial1

Use lowercase letters, numbers, and hyphens. Start with a letter.

Password *
.....  [Generate](#)

Set a password for the root user. [Learn more](#)

☐ No password

- ✓ Must contain an uppercase letter, lowercase letter, number and non-alphanumeric character
- ✓ Must not contain the username
- ✓ Must be at least 8 characters long

[Password policy](#)

Choose region and zonal availability

For better performance, keep your data close to the services that need it. Region is permanent, while zone can be changed any time.

Region
us-east1 (South Carolina)

Zonal availability

☒ Single zone

In case of outage, no failover. Not recommended for production.

☐ Multiple zones (Highly available)

Automatic failover to another zone within your selected region. Recommended for production instances. Increases cost.

b.

Customize your instance

You can also customize instance configurations later

Machine configuration



Machine

Choose the machine that will best fit the resource needs of your instance. Your machine selection determines the storage type and configuration options.

General purpose - Dedicated core



- ☐ 1 vCPU, 3.75 GB
- ☒ 2 vCPU, 8 GB
- ☐ 4 vCPU, 16 GB
- ☐ 8 vCPU, 32 GB
- ☐ Custom

Data cache

Exclusive to Enterprise Plus edition. Adds a local SSD for read caching to reduce read latency and improve performance at additional cost. [Learn more](#)

☐ Enable data cache

Storage



Storage type

- ☒ SSD
Most popular choice. Lower latency than HDD with higher QPS and data throughput.
- ☐ HDD
Lower performance than SSD with lower storage rates.

Storage capacity

10 - 65,536 GB. Higher capacity improves performance, up to the limits set by the machine type. Capacity can't be decreased later.

- ☒ 10 GB
- ☐ 20 GB
- ☐ 100 GB
- ☐ 250 GB
- ☐ Custom

☐ Enable automatic storage increases
If enabled, whenever you are nearing capacity, storage will be incrementally (and permanently) increased. [Learn more](#)

[Advanced encryption options](#)

Connections



Choose how you want your source to connect to this instance, then define which networks are authorized to connect. [Learn more](#)

You can use the Cloud SQL Proxy for extra security with either option. [Learn more](#)

Instance IP assignment

☐ Private IP
Assigns an internal, Google-hosted VPC IP address. Requires additional APIs and permissions. [Learn more](#)

☒ Public IP
Assigns an external, internet-accessible IP address. Requires using an authorized network or the Cloud SQL Proxy to connect to this instance. [Learn more](#)

d.

Authorized networks

You can specify CIDR ranges to allow IP addresses in those ranges to access your instance. [Learn more](#)

[Add a network](#)

Google Cloud services authorization

- ☐ **Enable private path**
Allows other Google Cloud services like BigQuery to access data and make queries over Private IP. [Learn more](#)

Security



SSL mode

Secure data transmission between client and this instance by enforcing SSL encryption and certificates. SSL encryption is recommended for all connections using public IP networks. [Learn more](#)

- ☐ **Allow unencrypted network traffic (not recommended)**
- ☒ **Allow only SSL connections**
Only allows connections using SSL/TLS encryption. Client certificates will not be verified.
- ☐ **Require trusted client certificates**
Only allows connections from clients that use a valid client certificate and SSL encryption

Server certificate authority mode

Choose the type of certificate authority that signs the server certificate for this Cloud SQL instance. [Learn more](#)

Server certificate authority mode can't be changed once chosen for the instance.

- ☐ **Google managed internal certificate authority**
An internal per-instance certificate authority is the trust anchor for the instance.
- ☒ **Google managed CAS certificate authority**
A root certificate authority and subordinate certificate authorities stored in Certificate Authority Service (CAS) are the trust anchors for all instances in a region.

e.

Data Protection

^

Backup tier

Choose the right backup tier based on the level of protection needed for your instance. Enhanced backup provides additional security for backups using Backup and DR service. Backup and DR service is a separate service from Cloud SQL with independent certifications and accreditation.

Compare Tiers

- ☒ **Standard backup tier**
Foundational protection with daily scheduling and PITR. [Learn more](#)

Automated backups and point-in-time recovery

Protect your data from loss at a minimal cost. Make sure your storage can support the automated backups and days of logs you're retaining. [Learn more](#)

- ☐ **Automated daily backups**
- ☐ **Enable point-in-time recovery**
Allows you to recover data from a specific point in time, down to a fraction of a second. Enables binary logs (required for replication).

- ☐ **Enhanced backup tier** New
Enhanced protection with flexible scheduling and backup vault storage. [Learn more](#)

Instance deletion protection

Safeguard against accidental deletion and data loss. [Learn more](#)

- ☐ **Prevent instance deletion**
Prevent accidental or unauthorized deletion of this instance. Disable this setting before attempting to delete
- ☐ **Retain backups after instance deletion**
Automated backups are retained based on your settings, while on-demand backups are kept until manually deleted. Storage is billed based on usage.
- ☐ **Final backup on instance deletion**
Final backups will be automatically created during the deletion of the instance. Final backup is stored for 30 days after deletion by default.

f.

Maintenance^

Maintenance typically only takes place once every few months, and requires your instance to be restarted while updates are made, which disrupts service briefly.

Maintenance timing ?

Choose the week to apply maintenance when a new version is available.

Any▼

Maintenance window

Choose the best day and time window for this instance to undergo routine maintenance.

Any window▼

Deny maintenance period

You can deny upcoming maintenance by creating a deny period, which can last up to 90 days. You can only have one deny period in effect at a time. [Learn more](#)

Start date📅End date📅

Maintenance will be denied starting at 12 AM EST

Maintenance will resume at 12 AM EST

☒ Repeat every year

[^ Advanced options](#)

g.

Flags and parameters

^

Flags allow you to customize granular aspects of your instance. Changes may require restart. [Learn more](#)

Add a database flag

Vertex AI Integration

This integrates Cloud SQL with Vertex AI, so you can apply the semantic and predictive power of machine learning (ML) models to your database. [Learn more](#)

- ☐ Enable Vertex AI Integration
- Allows this Cloud SQL instance to connect to Vertex AI and to send requests for real-time predictions and insights to large language models (LLMs).

Query insights

^

Query insights helps you detect and diagnose performance issues in your instance by examining queries using both historical and near real-time data. Enable it to understand database load, identify slow-running queries, view active queries, and gain visibility into apps that connect to this instance. **7-days telemetry at no additional cost.** [Learn more](#)

- ☐ Enable Query insights

- h.
- i. Double checking you have the same cost on the right panel:

Summary

Cloud SQL Edition	Enterprise
Region	us-east1 (South Carolina)
DB Version	MySQL 8.4
Machine type	db-custom-2-8192
vCPUs	2 vCPU
RAM	8 GB
Data Cache	Disabled
Storage	10 GB SSD
Connections	Public IP
Backup	Manual
Availability	Single zone
Point-in-time recovery	Disabled
Network throughput (MB/s)	500 of 500
IOPS	Read: 6,300 of 15,000 Write: 6,300 of 15,000
Disk throughput (MB/s)	Read: 4.8 of 240.0 Write: 4.8 of 240.0

Pricing estimate (without discounts)

These items represent Cloud SQL compute, memory and storage resources only, and reflect how you configured your instance so far. Discounts not included in estimate. [Learn more](#)

Item	Hourly cost (estimate)
2 vCPU (\$0.041 per vCPU/hour)	\$0.08
8 GiB RAM (\$0.007 per GiB/hour)	\$0.06
10 GiB SSD (\$0.17 per GiB/month)	\$0.002
Total without usage discounts	\$0.14

i.

9. At last, select “create instance,” it may take a few minutes to create and show up in your console.

Query insights

Query insights helps you detect and diagnose performance issues in your instance by examining queries using both historical and near real-time data. Enable it to understand database load, identify slow-running queries, view active queries, and gain visibility into apps that connect to this instance. 7-days telemetry at no additional cost. [Learn more](#)

☐ Enable Query insights

Labels

Labels can help you organize your Cloud SQL instances. [Learn more](#)

Add a label

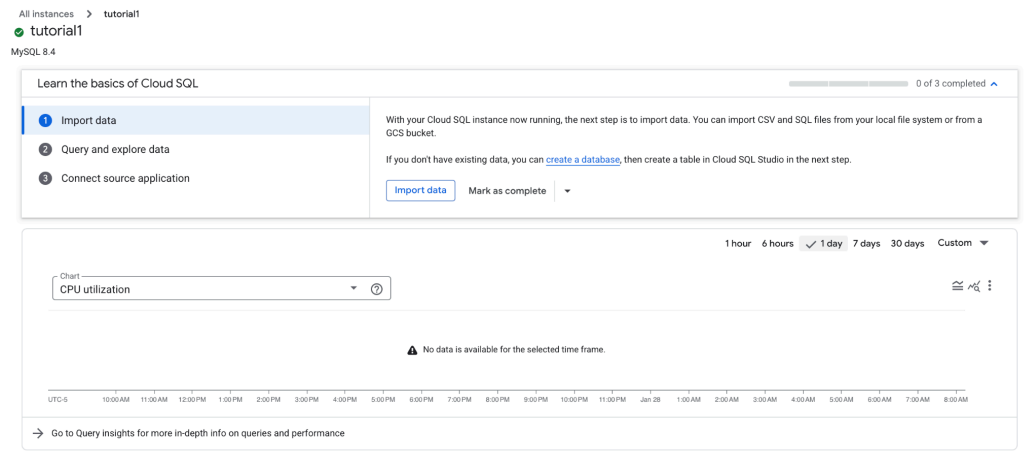
[^ Hide configuration options](#)

Create instance

Cancel

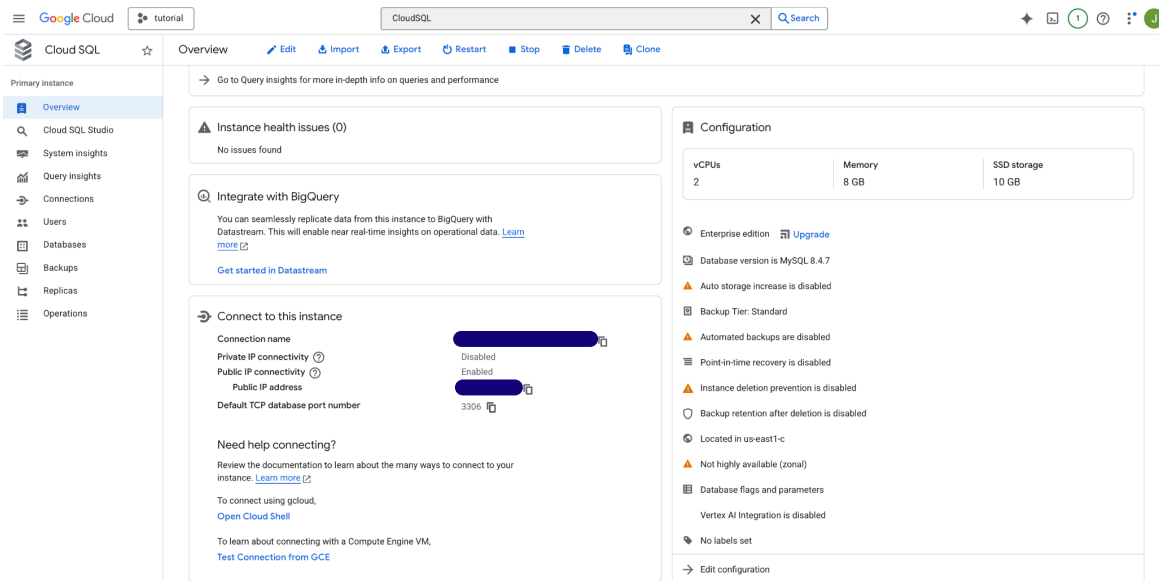
i.

10. After your instance has been successfully created, you should see the following:



a.

11. Scroll down the page. Go to the “Configuration” and select “edit configuration”



a.

12. Go down to “Connections” and select “add a network”

Connections

Choose how you want your source to connect to this instance, then define which networks are authorized to connect. [Learn more](#)

You can use the Cloud SQL Proxy for extra security with either option. [Learn more](#)

Instance IP assignment

☐

Private IP

Assigns an internal, Google-hosted VPC IP address. Requires additional APIs and permissions. [Learn more](#)

☒

Public IP

Assigns an external, internet-accessible IP address. Requires using an authorized network or the Cloud SQL Proxy to connect to this instance. [Learn more](#)

Authorized networks

You can specify CIDR ranges to allow IP addresses in those ranges to access your instance. [Learn more](#)

^
New network

Name
demo

IP range *

a.

- b. Give a name and your IP range. Then click “Done”
13. Go down the page and click “Save”
14. Now what we need to do next is to connect this CloudSQL instance you just set up to your local MySQL Workbench deployment. The information you need is in the “Connect to this instance” section

 **Connect to this instance**

Connection name		
Private IP connectivity 	Disabled	
Public IP connectivity 	Enabled	
Public IP address		
Default TCP database port number	3306 	

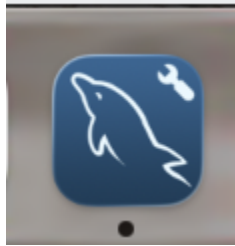
Need help connecting?

Review the documentation to learn about the many ways to connect to your instance. [Learn more](#) 

To connect using gcloud,
[Open Cloud Shell](#)

To learn about connecting with a Compute Engine VM,
[Test Connection from GCE](#)

- a.
15. [NOT on Google Cloud anymore] Open MySQL Workbench on your local computer



- a.
16. Click the + sign to add a connection

MySQL Connections  

 Filter connections

Local instance 3306

 root
 localhost:3306

- a.
17. Fill in the Connection Name with your personalized connection name that shows up in your “Connect to this instance” section of your CloudSQL setup

Connection Name: Type a name for the connection

Connection Method: Method to use to connect to the RDBMS

Parameters SSL Advanced

Hostname: Port: Name or IP address of the server host - and TCP/IP port.

Username: Name of the user to connect with.

Password: The user's password. Will be requested later if it's not set.

Default Schema: The schema to use as default schema. Leave blank to select it later.

a.

18. Then go to "Advanced"

Connection Name: Type a name for the connection

Connection Method: Method to use to connect to the RDBMS

Parameters SSL Advanced

Hostname: Port: Name or IP address of the server host - and TCP/IP port.

Username: Name of the user to connect with.

Password: The user's password. Will be requested later if it's not set.

Default Schema: The schema to use as default schema. Leave blank to select it later.

a.

19. Type "OPT_LOCAL_INFILE=1" in Others and then click "Test Connection" at the bottom.
If it works, then click OK

Setup New Connection

Connection Name: Type a name for the connection

Connection Method: Standard (TCP/IP) Method to use to connect to the RDBMS

Parameters | **SSL** | **Advanced**

☐ Use compression protocol Select this option for WAN connections.

☐ Use ANSI quotes to quote identifiers If enabled this option overwrites the serverside settings.

☐ Enable Cleartext Authentication Plugin Send user password in cleartext. Required for some authentication methods.

Timeout: Maximum time to wait before a connection is aborted.

SQL_MODE: Override the default SQL_MODE used by the server.

Others: Other options for Connector/C++ as option=value pairs, one per line.

a.

20. Then you should see something like this:

Setup New Connection

Connection Name: Type a name for the connection

Connection Method: Standard (TCP/IP) Method to use to connect to the RDBMS

Parameters | **SSL** | **Advanced**

Successfully made the MySQL connection

Information related to this connection:

Host: 127.0.0.1
Port: 3306
User: root
SSL: enabled with
TLS_AES_128_GCM_SHA256

A successful MySQL connection was made with the parameters defined for this connection.

a.

21. Now you should be able to see this connection under MySQL Connections now

Welcome to MySQL Workbench

MySQL Workbench is the official graphical user interface (GUI) tool for MySQL. It allows you to design, create and browse your database schemas, work with database objects and insert data as well as design and run SQL queries to work with stored data. You can also migrate schemas and data from other database vendors to your MySQL database.

[Browse Documentation >](#)

[Read the Blog >](#)

[Discuss on the Forums >](#)

MySQL Connections

 Filter connections

Local instance 3306

 root
 localhost:3306

tutorial-485...:us-east1:tutorial1

 root
 127.0.0.1:3306

a.